

Logic Exercises 2

1. Investigate which of the following semantic entailments hold:

- (a) $p \vee (q \rightarrow \neg p) \models p \vee \neg q$
- (b) $p \vee \neg q \models p \vee (q \rightarrow \neg p)$
- (c) $\models p \vee (q \rightarrow \neg p)$
- (d) $p \vee (q \rightarrow p) \models p \vee \neg q$
- (e) $p \vee \neg q \models p \vee (q \rightarrow p)$
- (f) $(p \rightarrow q) \rightarrow r \models p \rightarrow (q \rightarrow r)$
- (g) $p \rightarrow (q \rightarrow r) \models (p \rightarrow q) \rightarrow r$
- (h) $p \rightarrow (\neg q \vee r), \neg r \models q \rightarrow \neg p$

2. For each of the following specifications, try to find a formula ϕ which satisfies it. If no such ϕ exists, then explain why.

- (a) A formula ϕ such that $p \models \phi$ and $\neg p \models \phi$.
- (b) A formula ϕ such that $p \models \phi$ and $p \models \neg \phi$.
- (c) A formula ϕ such that $\phi \models p$ and $\phi \models \neg p$.

3. A binary relation $\mathcal{R}$ on a set S of elements is called:

- *reflexive* if $s \mathcal{R} s$ for all $s \in S$;
- *symmetric* if $s \mathcal{R} t$ implies $t \mathcal{R} s$, for all $s, t \in S$;
- *transitive* if $s \mathcal{R} t$ and $t \mathcal{R} u$ imply $s \mathcal{R} u$, for all $s, t, u \in S$.

- (a) Is the semantic entailment relation $\models$ reflexive? Is it symmetric? Is it transitive?
- (b) Is semantic equivalence $\equiv$ reflexive? Is it symmetric? Is it transitive?

4. Addition $+$ and multiplication $\cdot$ in arithmetic are (just as conjunction and disjunction in propositional logic) *commutative* and *associative*. Also a *distributivity* law holds for these two operators.

- (a) Write down the corresponding five laws for addition and multiplication: two commutative laws, two associative laws, and one distributivity law.
- (b) Which distributivity law for addition and multiplication does not hold?

1. Investigate which of the following semantic entailments hold:

(a) $p \vee (q \rightarrow \neg p) \models p \vee \neg q$

(b) $p \vee \neg q \models p \vee (q \rightarrow \neg p)$

(c) $\models p \vee (q \rightarrow \neg p)$ $\checkmark \rightarrow$ because is a tautology

(d) $p \vee (q \rightarrow p) \models p \vee \neg q$

(e) $p \vee \neg q \models p \vee (q \rightarrow p)$

(f) $(p \rightarrow q) \rightarrow r \models p \rightarrow (q \rightarrow r)$

(g) $p \rightarrow (q \rightarrow r) \models (p \rightarrow q) \rightarrow r$

(h) $p \rightarrow (\neg q \vee r), \neg r \models q \rightarrow \neg p$

a)

p	q	$q \rightarrow \neg p$	$p \vee (q \rightarrow \neg p)$	$p \vee \neg q$
T	T	F	T	T
T	F	T	T	T
F	T	T	T	F
F	F	T	T	T

b) $p \vee \neg q \models p \vee (q \rightarrow \neg p) \checkmark$

p	q	$q \rightarrow \neg p$	$p \vee (q \rightarrow \neg p)$	$p \vee \neg q$
T	T	F	T	T
T	F	T	T	T
F	T	T	T	F
F	F	T	T	T

$p \vee (q \rightarrow \neg p)$

for it to be F

$\nearrow p$

$\rightarrow (q \rightarrow \neg p)$

$q \rightarrow \text{True}$

$p - \text{true} \times$

$$p = T \quad q = T \Rightarrow p \vee (q \vee \neg p) \quad \text{always true}$$

T

$$f) (p \rightarrow q) \rightarrow r \quad \models \quad p \rightarrow (q \rightarrow r)$$

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \rightarrow r$	$p \rightarrow (q \rightarrow r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	F	T	T	T
F	T	F	F	F	T	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T



Semantic entailment holds

same table

$$(p \rightarrow q) \rightarrow r \models p \rightarrow (q \rightarrow r)$$

g)

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \rightarrow r$	$p \rightarrow (q \rightarrow r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	F	T	T	T
F	T	F	F	F	T	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

not valid

2. For each of the following specifications, try to find a formula ϕ which satisfies it. If no such ϕ exists, then explain why.

(a) A formula ϕ such that $p \models \phi$ and $\neg p \models \phi$.

(b) A formula ϕ such that $p \models \phi$ and $p \models \neg\phi$.

(c) A formula ϕ such that $\phi \models p$ and $\phi \models \neg p$.

↓
[pi]

a) ϕ s.t. $p \models \phi$ and $\neg p \models \phi$

$\begin{cases} p \models \phi \\ \neg p \models \phi \end{cases} \rightarrow$ either ① p is a contradiction
or if p is a tautology $\rightarrow \phi$ is also a tautology

②

① $p = \text{F}$ $p \models \phi$ ✓

for $\neg p \models \phi$ to hold $\rightarrow \phi$ must be a tautology

② $p = \text{T}$ $p \models \phi \rightarrow$ tautology

$\begin{array}{c} \neg p \models \phi \\ \uparrow \\ \text{F} \end{array} \quad \begin{array}{c} \uparrow \\ \text{tautology} \end{array} \quad \left| \rightarrow \text{always holds} \right.$

① $x \wedge \neg x \models y \vee \neg y$

② $x \vee \neg x \models z \vee \neg z$

b) (b) A formula ϕ such that $p \models \phi$ and $p \models \neg\phi$.

ϕ s.t. $p \models \phi$
and
 $p \models \neg\phi$

$$p \rightarrow \top$$

$$\left\langle \begin{array}{l} p \models \phi \\ p \models \neg \phi \end{array} \right| \Rightarrow p \models \phi \approx p \models \neg \phi$$

c) ϕ s.t. $\phi \models p$ and $\phi \models \neg p$

ϕ false $\rightarrow$ true
X

$$p \rightarrow \top \rightarrow \phi \text{ X} \rightarrow \phi \models \neg p \text{ does not hold}$$

$$p \rightarrow \text{F} \rightarrow \phi \text{ X} \rightarrow \phi \models p \text{ —// —}$$

It only holds when ϕ is a contradiction

$$\phi \equiv p \wedge \neg p$$

3. A binary relation $\mathcal{R}$ on a set S of elements is called:

- reflexive if $s \mathcal{R} s$ for all $s \in S$; $s \mathcal{R} s \quad \forall s \in S$
- symmetric if $s \mathcal{R} t$ implies $t \mathcal{R} s$, for all $s, t \in S$; $s \mathcal{R} t \rightarrow t \mathcal{R} s \quad \forall s, t \in S$
- transitive if $s \mathcal{R} t$ and $t \mathcal{R} u$ imply $s \mathcal{R} u$, for all $s, t, u \in S$. $s \mathcal{R} t \wedge t \mathcal{R} u \rightarrow s \mathcal{R} u \quad \forall s, t, u \in S$

(a) Is the semantic entailment relation $\models$ reflexive? Is it symmetric? Is it transitive?

(b) Is semantic equivalence $\equiv$ reflexive? Is it symmetric? Is it transitive?

a)

$$\begin{array}{l} \phi \quad \psi \quad \chi \\ \phi \quad \neg \top \\ \phi \models \end{array}$$

$\phi \equiv \phi$ ✓

$A \models B$ ✓ $B \models A$ ✓

$\phi \models \psi \wedge \psi \models \chi \rightarrow \phi \models \chi$ ✓

$\phi \models \psi \wedge \psi \models \chi \rightarrow \phi \models \chi$ ✓

4. Addition $+$ and multiplication $\cdot$ in arithmetic are (just as conjunction and disjunction in propositional logic) *commutative* and *associative*. Also a *distributivity* law holds for these two operators.

(a) Write down the corresponding five laws for addition and multiplication: two commutative laws, two associative laws, and one distributivity law.

(b) Which distributivity law for addition and multiplication does not hold?

$$\begin{array}{l} a) \quad k + l = l + k \\ \quad \quad k \cdot l = l \cdot k \end{array} \quad \left| \begin{array}{l} \text{commutativity} \end{array} \right.$$

$$\begin{array}{l} \text{associativity} \end{array} \quad \left| \begin{array}{l} (k + l) + m = k + (l + m) \\ (k \cdot l) \cdot m = k \cdot (l \cdot m) \end{array} \right.$$

$$\text{distributivity} \quad \left| \quad k \cdot (l + m) = k \cdot l + k \cdot m \right.$$

$$b) \quad (k \cdot l) + m = (k + m) \cdot (l + m) \quad \text{does NOT hold}$$

6. Let $(\phi \rightarrow \psi_1) \models \chi$.

(a) Suppose that $\psi_1 \models \psi_2$. Can we then conclude that $(\phi \rightarrow \psi_2) \models \chi$?

(b) Suppose that $\psi_2 \models \psi_1$. Can we then conclude that $(\phi \rightarrow \psi_2) \models \chi$?

No

$$6) \quad (\phi \rightarrow \psi_1) \models \chi$$

$$a) \quad \psi_1 \models \psi_2$$

$$(\phi \rightarrow \psi_1) \models \chi$$

$$\begin{array}{c} \underbrace{} \\ \downarrow \text{F} \\ \phi \rightarrow \text{T} \quad \psi_1 = \text{F} \end{array}$$

$$\begin{array}{l} \nearrow \psi_1 \text{ x} \\ \searrow \psi_1 = \text{T} \wedge \psi_2 = \text{T} \end{array}$$

$$\begin{array}{c} \uparrow \\ \text{F} \end{array}$$

$$(from task) \left| \begin{array}{ccc} \phi & \rightarrow & \gamma_1 \models \chi \\ \top & & \top \\ \text{ } & \underbrace{\quad \quad} & \\ & \text{ } & \top \end{array} \right. \quad \checkmark$$

$$\begin{array}{ccc} \gamma_1 & \models & \gamma_2 \\ \top & & \top \end{array} \quad \text{holds}$$

$$\gamma_2 \rightarrow \top \Rightarrow (\phi \rightarrow \gamma_2) \models \chi \quad \times$$

7.

$$\phi - \top$$

$$\gamma_1 - \top$$

$$\chi = \top$$

$$\gamma_2 - \top$$

$p \wedge \neg p \models q$ holds

extra
example

p	q	$p \wedge \neg p$	q
T	T	F	T
T	F	F	F
F	T	F	T
F	F	F	F

7. On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). During a walk on this island you meet islanders A and B , and ask whether the road to the harbour is straight ahead.

- A says: "Yes".
- B says: "No".
- "At least one of us is a liar", concludes B .

- Formalize these three claims in propositional logic.
- Determine for both A and B by means of a truth table whether this islander is a truth speaker or a liar.
- Is the road to the harbour straight ahead?

a) h : harbour is straight ahead
 t_A : A is a T speaker
 t_B : B is a T speaker

- $t_A \leftrightarrow h$
- $t_B \leftrightarrow \neg h$
- $t_B \leftrightarrow (\neg t_A \vee \neg t_B)$

b)

t_A	t_B	h	$t_A \leftrightarrow h$	$t_B \leftrightarrow \neg h$	$\neg t_A \vee \neg t_B$	$t_B \leftrightarrow (\neg t_A \vee \neg t_B)$
T	T	T	T	F	F	F
T	T	F	F	T	F	F
T	F	T	T	T	T	F
T	F	F	F	F	T	F
F	T	T	F	F	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	F
F	F	F	T	F	T	F



A $\rightarrow$ not truth speaker

B $\rightarrow$ truth speaker

c) $h \rightarrow F \rightarrow$ not straight ahead

OR

B $\rightarrow$ truth speaker $\rightarrow$ said no $\rightarrow$ not straight ahead

5. Which of the following statements are true? Prove if true, disprove if not true.
- (a) If $\phi \models \psi_1 \vee \psi_2$ holds, then $\phi \models \psi_1$ holds or $\phi \models \psi_2$ holds.
 - (b) If $\phi \models \psi_1 \wedge \psi_2$ holds, then $\phi \models \psi_1$ holds and $\phi \models \psi_2$ holds.
6. Let $(\phi \rightarrow \psi_1) \models \chi$.
- (a) Suppose that $\psi_1 \models \psi_2$. Can we then conclude that $(\phi \rightarrow \psi_2) \models \chi$?
 - (b) Suppose that $\psi_2 \models \psi_1$. Can we then conclude that $(\phi \rightarrow \psi_2) \models \chi$?
7. On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). During a walk on this island you meet islanders A and B , and ask whether the road to the harbour is straight ahead.
- A says: “Yes”.
 - B says: “No”.
 - “At least one of us is a liar”, concludes B .
- (a) Formalize these three claims in propositional logic.
 - (b) Determine for both A and B by means of a truth table whether this islander is a truth speaker or a liar.
 - (c) Is the road to the harbour straight ahead?

8. On the island of liars and truth speakers, you meet three islanders A , B and C .

- A says: “ B or C is a liar.”
- B says: “ C is a liar.”
- C says: “ A is a liar.”

$t_A: A$ is a truth speaker

$t_B: B$ is a truth speaker

$t_C: C$ is a truth speaker

Determine by means of logical reasoning which of these three islanders speak the truth and which ones lie. Also check explicitly that the claims by the three islanders are in line with your conclusion whether they are truth speakers or liars.

9. Suppose you ask the following question to an islander hurrying by on a T-crossing to find the way to the harbour:

Is the declarative sentence “ $you\ are\ a\ truth\ speaker \leftrightarrow the\ harbour\ is\ to\ the\ right$ ” true?

Either explain that this question suffices to determine the way to the harbour, or explain why this is not the case.